

01 Measurement

Power-Sharing Index construction and validation

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Loaded V-Dem Data

Raw file	V-Dem version	Rows	Columns
V-Dem-CY-Full+Others-v16.rds	v16	28092	4618

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Purpose

This file constructs the Power-Sharing Index (PSI) for the U.S. case using V-Dem data, checks basic measurement properties, compares aggregation choices, propagates V-Dem measurement uncertainty into the exported series, standardizes Polity2 as an optional external procedural-democracy comparator, and exports the derived PSI dataset for downstream analysis. The PDF is meant to be a lean replication document: figures render inside the PDF only, and publication graphics are generated separately when needed.

Setup

Load V-Dem data

Raw V-Dem data are expected in `data/raw/` and are not included in the GitHub repository. This file reads the local V-Dem file and records the source file, row count, and column count used for replication.

Filter U.S. case

The analysis uses the U.S. country-year series available in V-Dem from 1789 forward. Because the goal is to measure as much of American democratic development as the data allow, the core PSI prioritizes indicators with full coverage from 1789 rather than indicators that begin in 1900 and miss the first century of the series.

U.S. Analysis Sample

Country	First year	Last year	Country-years
United States	1789	2025	237

PSI Components

Dimension	Variable	Component	Interpretation
Political power-sharing	v2pepwsoc	Social Group Power	Political power distributed across social groups
Political power-sharing	v2pepwrgen	Gender Power	Political power distributed across gender
Equal civil standing	v2clsocgrp	Civil Liberties Equality	Civil liberties protected across social groups
Coercive restraint	v2cltort	Freedom from Torture	State restraint from torture
Coercive restraint	v2clkill	Freedom from Killings	State restraint from political killings

Define PSI components

PSI is constructed from five indicators that capture political power-sharing, equal civil standing, and coercive restraint.

Candidate Indicator Screen

Core PSI indicators prioritize full U.S. coverage from 1789

Variable	Indicator	First_Year	Coverage	Use	Rationale
v2pepwsoc	Social group power	1789	Full from 1789	Core PSI	Full historical coverage and conceptual fit
v2pepwrgen	Gender power	1789	Full from 1789	Core PSI	Full historical coverage and conceptual fit
v2peapssoc	Access to public services by social group	1900	Partial from 1900	Exclude: begins in 1900	Misses the first century of the U.S. series
v2clsocgrp	Civil liberties equality by social group	1789	Full from 1789	Core PSI	Full historical coverage and conceptual fit
v2cltort	Freedom from torture	1789	Full from 1789	Core PSI	Full historical coverage and conceptual fit
v2clkill	Freedom from political killings	1789	Full from 1789	Core PSI	Full historical coverage and conceptual fit
v2x_suffr	Suffrage	1789	Full from 1789	Exclude: procedural suffrage	Captures electoral inclusion within the recognized demos rather than group
v2clpolcl	Political civil liberties	1900	Partial from 1900	Exclude: begins in 1900	Misses the first century of the U.S. series

Variable selection checks

The core selection constraint is historical coverage: the U.S. series begins in 1789, so core PSI indicators should cover the full available period. Indicators that begin in 1900 may be useful for supplementary analyses, but they are not used in the core measure because they miss the first century of American democratic development.

Constructed PSI Specifications

Statistic	Value
PSI additive min	0.000
PSI additive max	0.965
PSI geometric min	0.010
PSI geometric max	0.964
PSI gated min	0.000
PSI gated max	0.924
PSI hybrid min	0.003
PSI hybrid max	0.872

Polity2 Standardization Check

External comparator exported only for robustness checks

Source	Coverage	Usable country-years	Scale
e_polity2	1789-2018	230	Polity2 rescaled from -10 to 10 onto 0 to 1

Construct PSI

The five core indicators are normalized to a 0 to 1 scale within the U.S. historical series. Higher values indicate greater democratic power-sharing, and three aggregation rules are retained so downstream analysis can distinguish the transparent additive score from more restrictive geometric and hybrid specifications.

The geometric specification uses a small floor (0.01) to prevent a single component at its historical minimum from mechanically reducing the full product to zero. The gated PSI separates the core power-sharing dimension from coercive restraint and multiplies them, providing a theory-driven robustness check.

This diagnostic checks whether limited movement in the First Opening period reflects the underlying V-Dem source indicators or the PSI construction pipeline. If the raw indicators are piecewise-constant, the flatness is a data-resolution feature rather than a normalization artifact.

Aggregation sensitivity

This section compares additive, geometric, and hybrid aggregation rules. The purpose is to assess whether the substantive PSI trajectory depends on a particular formula or remains stable across reasonable specifications.

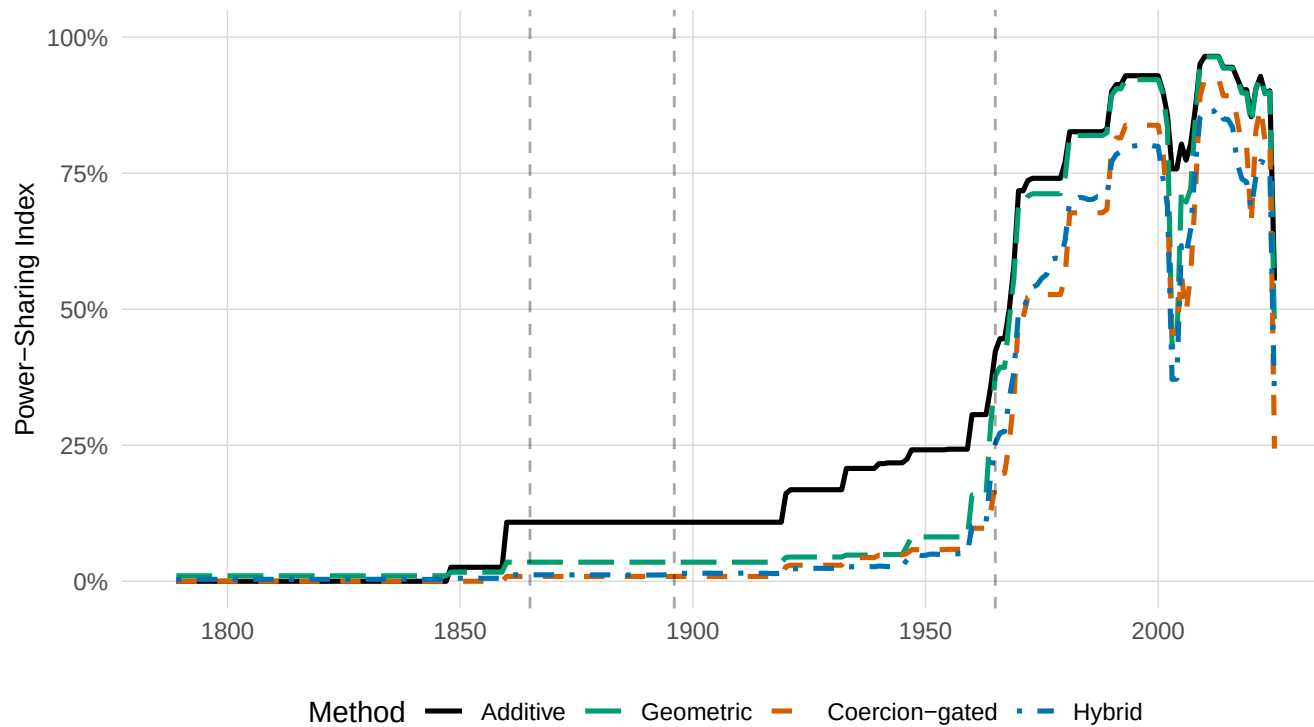


Figure 1: PSI by aggregation method.

Raw Component Diagnostic, 1850-1900

Unnormalized V-Dem indicators before PSI construction

year	v2pepwsoc	v2pepwrge	v2clsocgrp	v2cltort	v2clkill
1850	-0.341	-1.564	-2.136	-0.271	0.968
1851	-0.341	-1.564	-2.136	-0.271	0.968
1852	-0.341	-1.564	-2.136	-0.271	0.968
1853	-0.341	-1.564	-2.136	-0.271	0.968
1854	-0.341	-1.564	-2.136	-0.271	0.968
1855	-0.341	-1.564	-2.136	-0.271	0.968
1856	-0.341	-1.564	-2.136	-0.271	0.968
1857	-0.341	-1.564	-2.136	-0.271	0.968
1858	-0.341	-1.564	-2.136	-0.271	0.968
1859	-0.341	-1.564	-2.136	-0.271	0.968
1860	-0.341	-1.564	-2.136	0.756	0.968
1861	-0.341	-1.564	-2.136	0.756	0.968
1862	-0.341	-1.564	-2.136	0.756	0.968
1863	-0.341	-1.564	-2.136	0.756	0.968
1864	-0.341	-1.564	-2.136	0.756	0.968
1865	-0.341	-1.564	-2.136	0.756	0.968
1866	-0.341	-1.564	-2.136	0.756	0.968
1867	-0.341	-1.564	-2.136	0.756	0.968
1868	-0.341	-1.564	-2.136	0.756	0.968
1869	-0.341	-1.564	-2.136	0.756	0.968
1870	-0.341	-1.564	-2.136	0.756	0.968
1871	-0.341	-1.564	-2.136	0.756	0.968
1872	-0.341	-1.564	-2.136	0.756	0.968
1873	-0.341	-1.564	-2.136	0.756	0.968
1874	-0.341	-1.564	-2.136	0.756	0.968
1875	-0.341	-1.564	-2.136	0.756	0.968
1876	-0.341	-1.564	-2.136	0.756	0.968
1877	-0.341	-1.564	-2.136	0.756	0.968
1878	-0.341	-1.564	-2.136	0.756	0.968
1879	-0.341	-1.564	-2.136	0.756	0.968
1880	-0.341	-1.564	-2.136	0.756	0.968
1881	-0.341	-1.564	-2.136	0.756	0.968
1882	-0.341	-1.564	-2.136	0.756	0.968
1883	-0.341	-1.564	-2.136	0.756	0.968
1884	-0.341	-1.564	-2.136	0.756	0.968
1885	-0.341	-1.564	-2.136	0.756	0.968
1886	-0.341	-1.564	-2.136	0.756	0.968
1887	-0.341	-1.564	-2.136	0.756	0.968
1888	-0.341	-1.564	-2.136	0.756	0.968
1889	-0.341	-1.564	-2.136	0.756	0.968
1890	-0.341	-1.564	-2.136	0.756	0.968
1891	-0.341	-1.564	-2.136	0.756	0.968
1892	-0.341	-1.564	-2.136	0.756	0.968
1893	-0.341	-1.564	-2.136	0.756	0.968
1894	-0.341	-1.564	-2.136	0.756	0.968
1895	-0.341	-1.564	-2.136	0.756	0.968
1896	-0.341	-1.564	-2.136	0.756	0.968
1897	-0.341	-1.564	-2.136	0.756	0.968
1898	-0.341	-1.564	-2.136	0.756	0.968
1899	-0.341	-1.564	-2.136	0.756	0.968
1900	-0.341	-1.564	-2.136	0.756	0.968

PSI by Era and Aggregation Method

Era means, 0 to 1 scale

era	N	Additive mean	Geometric mean	Gated mean	Hybrid mean
Foundational Exclusion (1789-1865)	77	0.012	0.013	0.001	0.004
First Opening for Multiracial Democracy (1866-1896)	31	0.108	0.035	0.009	0.012
Partial Inclusion and Retrenchment (1897-1964)	68	0.182	0.060	0.037	0.034
Second Opening for Multiracial Democracy (1965-present)	61	0.821	0.792	0.669	0.667

Correlation Across Aggregation Methods

Comparison	Correlation
Additive vs. Geometric	0.983
Additive vs. Gated	0.975
Additive vs. Hybrid	0.978
Geometric vs. Hybrid	0.998
Gated vs. Hybrid	0.996

Component sensitivity

This check recalculates the additive PSI after dropping each component in turn. If the overall pattern remains stable, the measure is less likely to be an artifact of any single indicator.

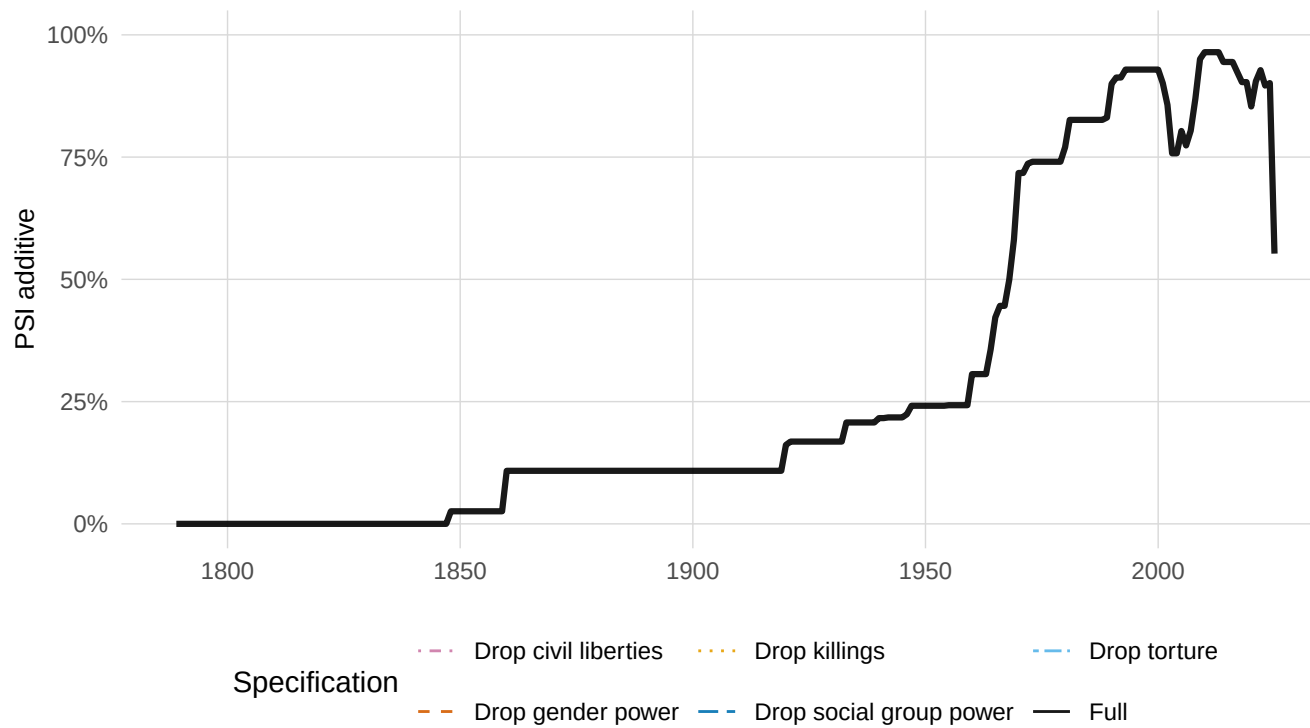


Figure 2: Leave-one-component-out sensitivity check.

The overlapping lines are substantively meaningful: they show that no single component drives the historical trajectory. For a publication appendix, this diagnostic can be replaced with a more compact summary table if space is limited.

Alternative specifications

Additional specifications test stronger and weaker assumptions about compensation across dimensions. The goal is not to select every possible formula, but to show that the core historical pattern is not fragile.

PSI Response to Historical Events

Additive specification, before-after comparisons

Event	Year_Before	Year_After	Expected	PSI_Before	PSI_After	Change	Direction
19th Amendment	1919	1921	Increase	0.108	0.168	0.060	Increase
Voting Rights Act	1964	1966	Increase	0.356	0.446	0.089	Increase
Shelby County v. Holder	2012	2014	Decrease	0.965	0.945	-0.020	Decrease

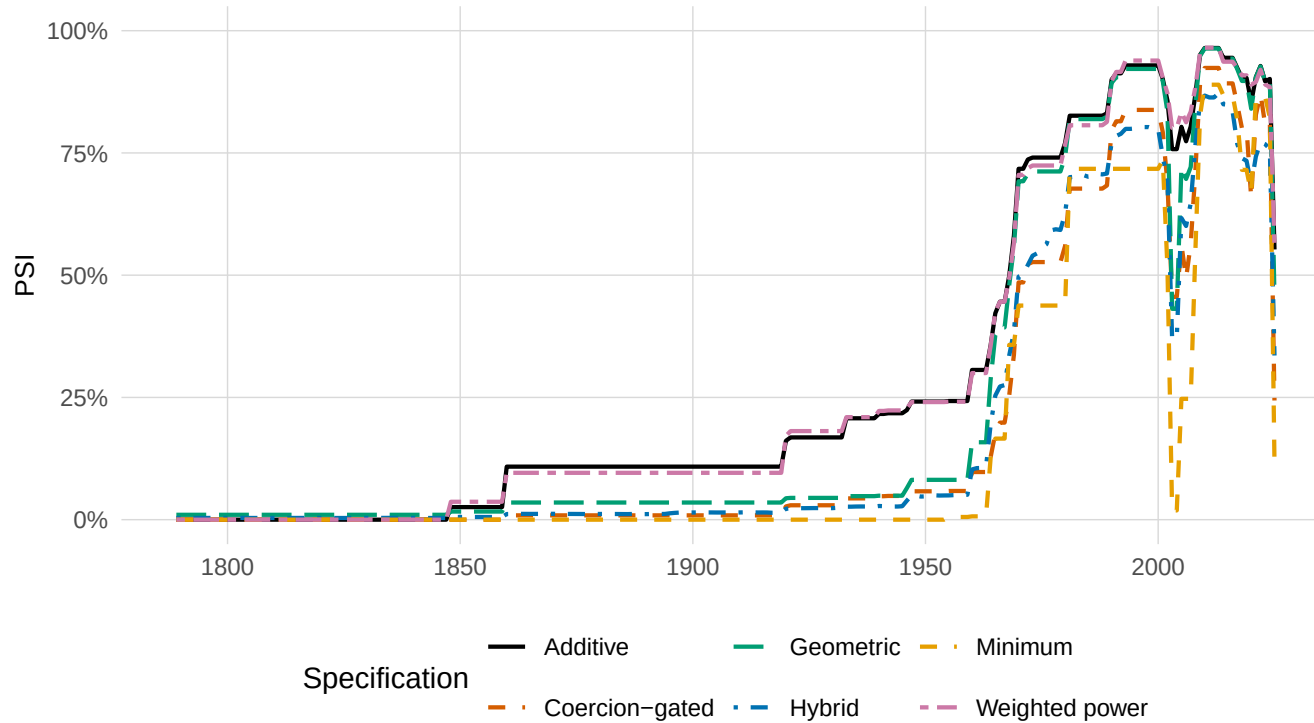


Figure 3: Alternative PSI specifications.

Known-event response

Known-event checks assess whether the PSI moves in expected directions around major institutional changes. These are descriptive validity checks, not causal estimates.

Internal Consistency

Metric	Value
Raw alpha	0.962
Standardized alpha	0.963
Average inter-item r	0.840

Internal consistency

Internal consistency checks whether the five normalized components move together as a coherent empirical scale. The full correlation matrix is shown so all five indicators remain visible; reliability is interpreted alongside the theoretical justification and sensitivity checks.

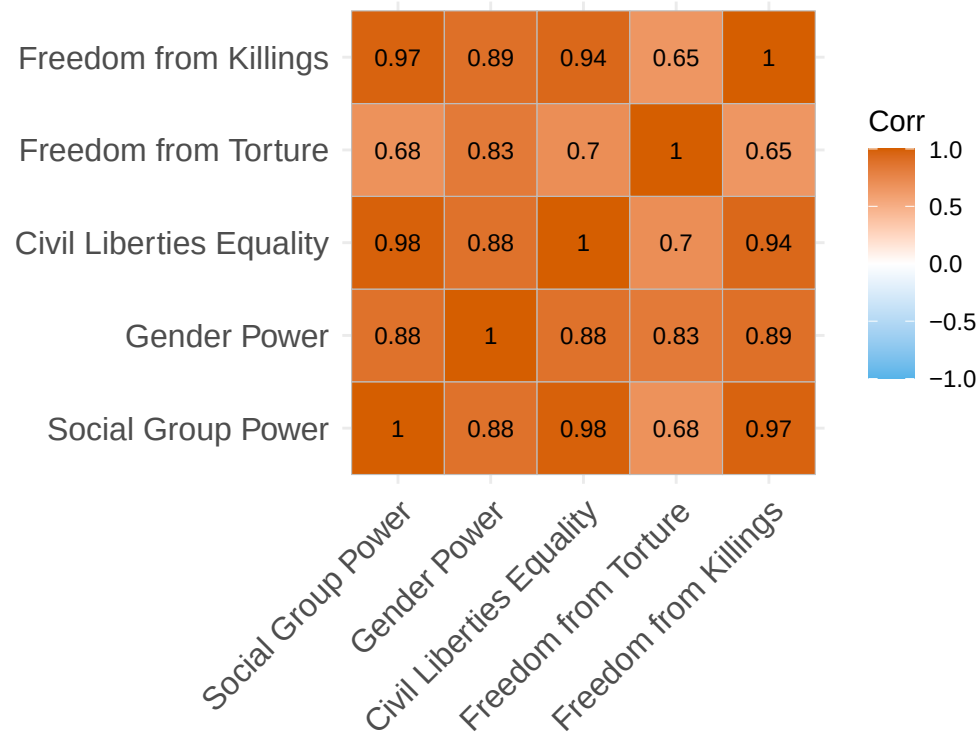


Figure 4: Inter-item correlations among normalized PSI components.

Inter-item correlations among normalized PSI components.

Alpha if Item Dropped

Component	raw_alpha	std.alpha
Social Group Power	0.943	0.946
Gender Power	0.948	0.948
Civil Liberties Equality	0.946	0.947
Freedom from Torture	0.977	0.979
Freedom from Killings	0.948	0.950

PCA Variance Explained

PC	Variance explained	Cumulative variance
PC1	87.5	87.5
PC2	9.4	96.9
PC3	2.0	98.9
PC4	0.9	99.8
PC5	0.2	100.0

Dimensionality

Dimensionality checks whether the indicators share a dominant empirical dimension. PCA and KMO results are used as diagnostics, not as substitutes for the theoretical definition of PSI.

PCA scree plot for PSI components.

PCA Loadings

Component	PC1	PC2
Social Group Power	0.464	0.308
Gender Power	0.458	-0.207
Civil Liberties Equality	0.463	0.254
Freedom from Torture	0.391	-0.825
Freedom from Killings	0.457	0.343

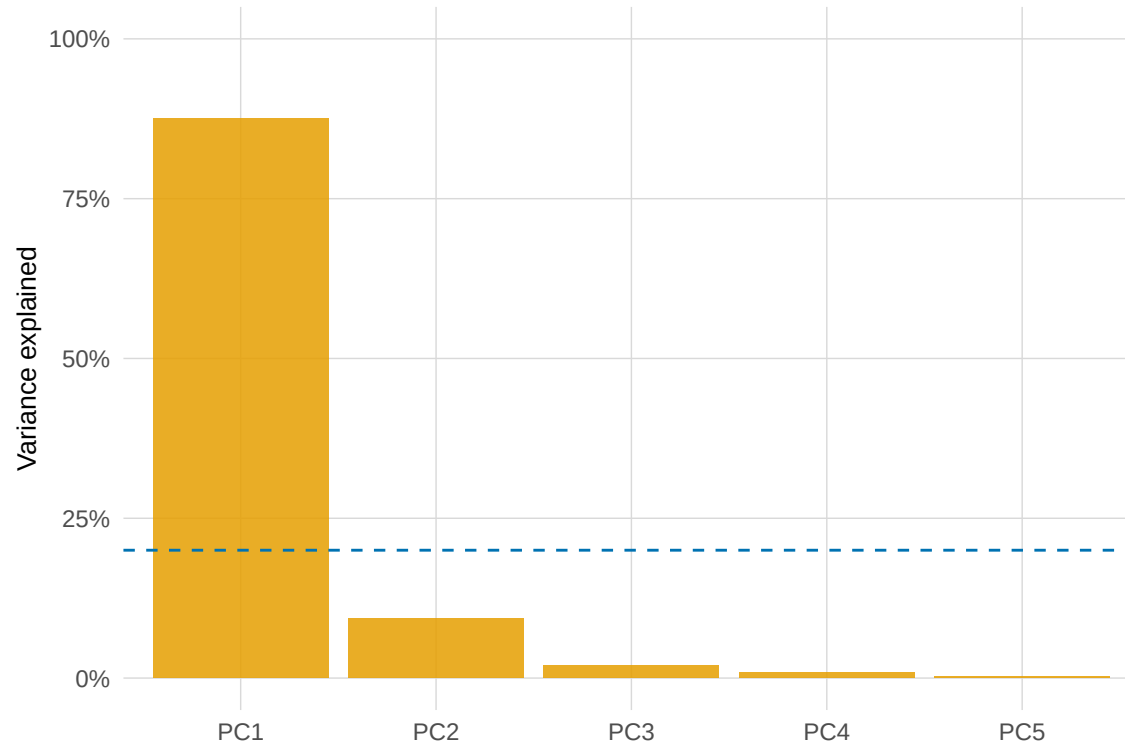


Figure 5: PCA scree plot for PSI components.

Kaiser-Meyer-Olkin Check

Item	KMO
Overall MSA	0.787
Social Group Power	0.747
Gender Power	0.806
Civil Liberties Equality	0.807
Freedom from Torture	0.768
Freedom from Killings	0.808

Era-Specific Correlations

PSI compared with V-Dem democracy indices and Polity2 when available

era	N	r(PSI, Electoral)	r(PSI, Liberal)	r(PSI, Participatory)	r(PSI, Polity2)
Foundational Exclusion (1789-1865)	77	-0.526	0.211	0.552	-0.358
First Opening for Multiracial Democracy (1866-1896)	31	NA	NA	NA	NA
Partial Inclusion and Retrenchment (1897-1964)	68	0.975	0.979	0.984	NA
Second Opening for Multiracial Democracy (1965-present)	61	0.858	0.863	0.864	0.472

V-Dem and external comparison checks

PSI is compared with major V-Dem democracy indices and, when available, the standardized Polity2 score. The V-Dem indices share V-Dem's measurement architecture, so agreement with them is partly mechanical. Polity2 is coded by a separate project, so divergence from Polity2 is useful as external discriminant evidence. Era-specific correlations are especially useful because the theory expects divergence during periods when democratic procedures existed within restricted membership.

Temporal stability

These intervals summarize variation across years within eras. They are not V-Dem measurement uncertainty intervals.

Full-Series Correlations

Index	Correlation
Electoral (V-Dem)	0.959
Liberal (V-Dem)	0.979
Participatory (V-Dem)	0.975
Polity2 (external)	0.399

Temporal Stability by Era

Within-era dispersion of the additive PSI

era	N	Mean	SD	SE	CI_Lower	CI_Upper
Foundational Exclusion (1789-1865)	77	0.012	0.030	0.003	0.006	0.019
First Opening for Multiracial Democracy (1866-1896)	31	0.108	0.000	0.000	0.108	0.108
Partial Inclusion and Retrenchment (1897-1964)	68	0.182	0.065	0.008	0.167	0.197
Second Opening for Multiracial Democracy (1965-present)	61	0.821	0.135	0.017	0.787	0.854

Summary of Measurement Evidence

Check	Evidence
Temporal coverage	Final components available from 1789
Aggregation sensitivity	Additive, geometric, and hybrid specifications preserve the same pattern
Component sensitivity	Leave-one-component-out checks preserve the same pattern
Internal consistency	Raw alpha = 0.962
Dimensionality	PC1 explains 87.5% of variance
Known-event response	PSI moves in expected directions around major historical events
V-Dem and external comparison	Era-specific correlations distinguish PSI from procedural democracy; Polity2 is retained as an independent comparator

Validation summary

The summary below reports the main measurement evidence in compact form. It is intended as a replication checkpoint rather than a claim that any single statistic fully validates the measure.

Saved Output

Output	Contents
data/derived/usa_psi_final.rda	Primary R data object for downstream QMDs

Export final PSI data

The final derived object is saved once and used by downstream QMDs. This keeps the workflow modular: `01_measurement.qmd` builds the measure, while later files use the completed PSI dataset. The standardized `polity2_01` variable is carried forward only as a procedural-democracy robustness comparator.

Session info

R version 4.5.3 (2026-03-11)

Platform: aarch64-apple-darwin20

Running under: macOS Tahoe 26.5.2

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1

locale:

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time zone: America/Indiana/Indianapolis

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attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

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loaded via a namespace (and not attached):

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